DUAL-RELATION REGIONAL MULTI-SCALE GRAPH FORECASTING FOR SPATIAL TIME SERIES WITH MISSING DATA

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ABSTRACT

Spatial monitoring streams rely on node histories and inter-sensor relations. Asynchronous missing observations across sensors create two coupled challenges: they shorten the available history at individual nodes and reduce the pairwise co-observations used to infer relations for graph message passing. Existing missing-data forecasting methods primarily handle incomplete histories through imputation or node-wise masks, without explicitly assessing whether learned relations are supported by sufficient pairwise evidence. We propose Dual-Relation Regional Multi-Scale Graph Forecasting (DRR-MSGF), a graph forecasting framework that directly predicts future values from incomplete spatiotemporal histories. Before sparse selection, DRR-MSGF regularizes window-specific learned relation scores with pairwise co-observation, fuses the resulting graph with physical topology, and aggregates regional context within physically connected regions. Missingness and regional features then gate regional-message injection and bias the multi-scale readout. Compared with the graph-based baseline HD-TTS-AMP, DRR-MSGF lowers MAE in all five missingness settings on METR-LA traffic-speed data and also when a high-rate Point mask is additionally applied to AQI36 air-quality data.

***Index Terms*—** Multivariate time series forecasting, missing observations, spatiotemporal graph learning, graph coarsening

1. INTRODUCTION

Traffic, environmental, and industrial forecasting systems depend on temporal context and inter-sensor relations. Spatiotemporal graph models represent these relations with physical topology or learned adjacency [1, 2], while hybrid designs combine physical and data-driven dependencies [3]. Both relation types are used to propagate information across sensors. Under asynchronous outages, similar node representations may be supported by markedly different amounts of joint observation. Missingness therefore affects not only node histories, but also whether an inferred edge has sufficient evidence for message passing.

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Missing-data research primarily addresses the node-history problem. Imputation reconstructs values through bidirectional recurrent models [4], temporal attention for irregular samples [5], or graph propagation [6, 7]; direct forecasting instead encodes values and masks jointly [8, 9]. These routes make incomplete histories usable for prediction. Once their features enter graph learning, however, node availability alone does not state how much jointly observed history supports a candidate node pair.

Pairwise evidence cannot be recovered from a node-wise missingness indicator. Comparable adaptive similarity scores may be estimated from only a few coincident observations or from a long jointly observed history; the score reflects numerical affinity, but not the amount of evidence supporting that affinity. The distinction matters before top-k selection commits candidate relations to subsequent graph propagation.

Relation modeling under irregular observation has also been studied beyond forecasting. RAINDROP learns sample-specific sensor graphs from irregular observations for classification [10], while TGNN4I handles partial node observations and irregular times through continuous-time graph states [11]. For missing-data forecasting, BiTGraph injects missing-pattern biases into temporal and graph propagation [12]; GinAR combines predefined and adaptive graphs under variable missingness [13]; and GCN-M studies graph learning with incomplete traffic observations [14]. To our knowledge, these methods jointly model missingness with node representations or graph construction, but do not use pairwise co-observation as explicit support for learned relations before sparse selection.

Multi-scale models further aggregate context across time and space [15, 16, 17]. A physical topology supplies a stable locality prior that does not depend on the current missing window, whereas a learned graph can capture window-specific functional dependence. They are complementary, but neither source alone determines whether a coarse regional message is useful for a particular node. A connected region can still have sparse coverage, long outages, or a node representation far from its regional center. Aggregating the fused graph within physically connected regions defined by the physical topology preserves both sources of relational evidence in the regional context; conditioning that context on node, observa-

tion, and regional cues avoids treating coarse aggregation as universally beneficial.

This raises a specific question: given an incomplete history window, how should a model condition learned-edge propagation and induced regional messages on available observation support? The goal is not to recover a ground-truth graph or calibrated uncertainty, but to condition forecasting messages on available evidence. DRR-MSGF regularizes learned relations with co-observation, fuses them with physical topology, aggregates the fused graph within physically connected regions, and uses observation and regional features to gate regional-message injection and bias multi-scale readout. Our contributions are:

- We develop observation–physical relation fusion for incomplete histories. A window-specific observation graph is constructed from mask-aware representations, with pairwise co-observation conditioning candidate edges before sparse selection, and fused edge-wise with physical topology.
- We introduce fusion-before-regionalization, aggregating the fused graph within fixed physically connected regions to construct regional context that retains both relation sources.
- We develop observation- and region-conditioned control of regional context, modulating regional-message injection and biasing multi-scale readout.

2. METHOD

2.1. Problem Setup and Backbone

Let $X \in \mathbb{R}^{T\times N\times C}$ be a history window, $M \in \{0,1\}^{T\times N\times C}$ its observation mask, and $A^{\mathrm{phy}} \in \mathbb{R}^{N\times N}$ the physical adjacency. The model maps $X^{\mathrm{obs}} = M \odot X$, M, and A^{phy} to $\hat{Y} \in \mathbb{R}^{H\times N\times C_y}$ for future target Y of the same shape. Graphs and observation features are computed per window; we omit the batch index. All datasets have one target channel per node ($C = C_y = 1$), so $\bar{M}_{t,i} = M_{t,i,1}$ and

$$r_i = \frac{1}{T}\sum_t \bar{M}_{t,i}, \qquad q_{ij} = \frac{1}{T}\sum_t \bar{M}_{t,i}\bar{M}_{t,j}. \tag{1}$$

Here, r_i is availability, q_{ij} is pairwise co-observation support, and $\epsilon > 0$ avoids division by zero and $\log 0$; propagation adjacencies are row-normalized. We retain the mask-aware encoder, dilated temporal hierarchy, fixed physical spatial hierarchy, and readout of HD-TTS-AMP [15]. It maps incomplete history to level-wise node states $H_\ell \in \mathbb{R}^{N\times d}$ and multi-resolution spatial states $B_{s,\ell}$, where ℓ indexes temporal levels and $s = 0$ the finest spatial level. DRR-MSGF replaces only $B_{0,\ell}$ by $\widetilde{B}_{0,\ell}$; $B_{s,\ell}$ for $s > 0$ remains complementary readout context. All $W_q, W_k, W_f, W_s, W_c, W_r, W_p \in \mathbb{R}^{d\times d}$ and $w_{s,\ell} \in \mathbb{R}^4$.

Fig. 1 outlines DRR-MSGF: co-observation guides fine relations, which are aggregated over physical regions and controlled by missingness-conditioned features.

2.2. Observation-Supported Relation Fusion

The encoder produces history features $E \in \mathbb{R}^{T\times N\times d}$. We obtain a window-specific node representation by masked pooling:

$$u_i = \frac{\sum_t \bar{M}_{t,i}E_{t,i}}{\sum_t \bar{M}_{t,i} + \epsilon}. \tag{2}$$

The observation-supported score combines learned similarity and co-observation:

$$s_{ij} = \frac{(W_q u_i)^\top (W_k u_j)}{\sqrt{d}}, \tag{3}$$

$$\tilde{s}_{ij} = s_{ij} + \mathrm{softplus}(\beta)\log(q_{ij} + \epsilon). \tag{4}$$

Here β is learnable. We retain the top-k entries of each directed score row ($k = 8$), apply row softmax, add an identity contribution, and row-normalize to obtain sparse A^{obs}. A selected diagonal score is learned, whereas the added identity is the only fixed self-loop. If $r_i = 0$, we set $A^{\mathrm{obs}}_{i,:} = 0$ to prevent arbitrary top-k ties. The logarithmic term softly discourages little-supported edges before sparsification.

Let $\bar{A}^{\mathrm{phy}}$ denote the physical graph after adding self-loops and applying row normalization. A learned edge gate then forms the fine relation:

$$g_{ij} = \mathrm{sigmoid}\Big(\psi_g([r_i, r_j, q_{ij}, \bar{A}^{\mathrm{phy}}_{ij}, A^{\mathrm{obs}}_{ij}])\Big), \tag{5}$$

$$A^{\mathrm{fine}} = \mathrm{RowNorm}\big(g \odot \bar{A}^{\mathrm{phy}} + (1-g)\odot A^{\mathrm{obs}}\big). \tag{6}$$

After Eq. (6), we overwrite $A^{\mathrm{fine}}_{i,:}$ with $\bar{A}^{\mathrm{phy}}_{i,:}$ whenever $r_i = 0$. Thus, both physical structure and observation-supported relations participate in end-to-end learned relation fusion. The gate is intentionally edge-specific rather than a global mixing coefficient, allowing physical and observation-supported evidence to receive different weights for different edges. In implementation, ψ_g is a two-layer $5 \to 16 \to 1$ MLP with GELU activation.

2.3. Missingness-Conditioned Regional Context

We construct a fixed hard assignment $S \in \{0,1\}^{N\times K}$ from the physical graph, assigning each node to one region. METR-LA uses distance connectivity thresholded at 0.1; AQI36 uses Haversine Gaussian weights $\exp[-(d_{ij}/\sigma)^2] \geq 0.1$, where σ is the standard deviation of nonzero distances. Both are directed and row-normalized for propagation. Only for region construction, we symmetrize the graph and remove self-loops. We seed by remaining physical degree, then greedily add the unassigned node with largest positive affinity until none remains or size $m = 8$ is reached. Thus

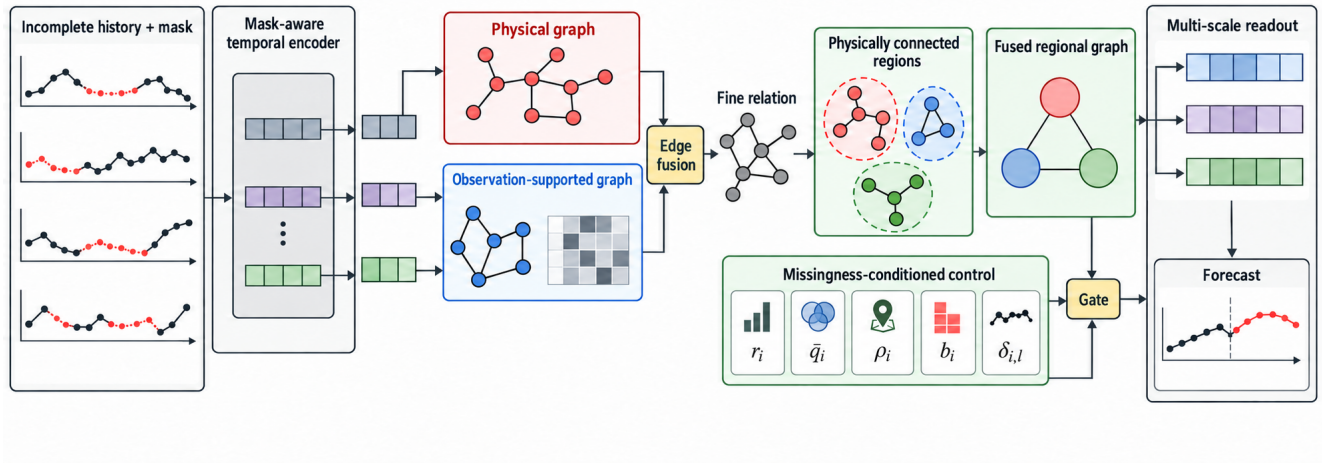


Fig. 1. Overview of DRR-MSGF. A mask-aware encoder forms temporal states from incomplete history. The fused graph is aggregated over connected regions defined solely by the physical graph. Observation and regional features regulate regional injection and multi-scale readout.

regions are connected and exclude zero-affinity nodes. With $n_k = \sum_i S_{ik}$,

$$Z_\ell^{\text{fine}} = \text{ELU}(A^{\text{fine}} H_\ell W_f + H_\ell W_s), \tag{7}$$

$$H_{\ell,k}^{\text{reg}} = \frac{1}{n_k} \sum_i S_{ik} Z_{\ell,i}^{\text{fine}}, \tag{8}$$

$$A^{\text{coarse}} = \text{RowNorm}(S^\top A^{\text{fine}} S). \tag{9}$$

Regional propagation and lifting are

$$Z_\ell^{\text{reg}} = \text{ELU}(A^{\text{coarse}} H_\ell^{\text{reg}} W_c + H_\ell^{\text{reg}} W_r), \tag{10}$$

$$Z_\ell^{\text{coarse}} = S Z_\ell^{\text{reg}}. \tag{11}$$

Coarsening follows fusion, so regional aggregation retains both physical and observation-supported relations. $S^\top A^{\text{fine}} S$ sums fine-edge weights: A^{coarse} is total inter-region edge mass rather than average affinity, and row normalization yields outgoing propagation weights.

For node i, let $\bar{q}_i = \sum_j \bar{A}_{ij}^{\text{phy}} q_{ij}$ be physical-neighbor-weighted co-observation, $\rho_i = \sum_k S_{ik} n_k^{-1} \sum_j S_{jk} r_j$ be regional coverage, and b_i be the longest missing block in the current history window divided by T. Let $k(i)$ denote the region of node i, $c_{i,\ell} = H_{\ell,k(i)}^{\text{reg}}$, and

$$\zeta_{i,\ell} = \sqrt{\max\left(d^{-1} \| Z_{\ell,i}^{\text{fine}} - c_{i,\ell} \|_2^2, \epsilon\right)}, \tag{12}$$

$$\delta_{i,\ell} = \frac{\zeta_{i,\ell}}{N^{-1} \sum_j \zeta_{j,\ell} + \epsilon}. \tag{13}$$

Here $\delta_{i,\ell}$ is normalized representation dispersion from the regional center. The missingness-conditioned regional gate is

$$\alpha_{i,\ell} = \text{sigmoid}\left(\psi_\alpha([r_i, \bar{q}_i, \rho_i, b_i, \delta_{i,\ell}]) \right. \tag{14}$$

$$\left. + \gamma^\top [1 - r_i, 1 - \bar{q}_i, \rho_i - 1, b_i, -\delta_{i,\ell}]\right). \tag{15}$$

Here ψ_α is the same two-layer MLP as the edge gate and γ is learnable. Its signed linear residual is not interpreted as a monotonic prior. Let $\boldsymbol{\alpha}_\ell = [\alpha_{1,\ell}, \ldots, \alpha_{N,\ell}]^\top \in \mathbb{R}^N$, broadcast along features. The finest representation becomes

$$\widetilde{B}_{0,\ell} = Z_\ell^{\text{fine}} + \boldsymbol{\alpha}_\ell \odot \left(Z_\ell^{\text{coarse}} W_p\right), \tag{16}$$

while $\widetilde{B}_{s,\ell} = B_{s,\ell}$ for $s > 0$. We further bias multi-scale readout by observation and regional features. Let $R_i = [r_i, \bar{q}_i, \rho_i, b_i]^\top$. The score for prediction step h is

$$e_{h,s,\ell,i} = e_{h,s,\ell,i}^{\text{repr}} + w_{s,\ell}^\top R_i. \tag{17}$$

$w_{s,\ell}$ is shared across prediction steps; horizon dependence enters through $e_{h,s,\ell,i}^{\text{repr}}$. A softmax over (s,ℓ) yields $\pi_{h,s,\ell,i}$ and $F_{h,i} = \sum_{s,\ell} \pi_{h,s,\ell,i} \widetilde{B}_{s,\ell,i}$, which the prediction head maps to $\hat{Y}_{h,i}$. Thus, these features bias scale selection rather than replace representations. Training minimizes masked forecasting MAE,

$$\mathcal{L}_{\text{pred}} = \frac{\sum M^Y \odot |\hat{Y} - Y|}{\sum M^Y}, \tag{18}$$

where M^Y excludes unavailable targets; no reconstruction, graph, or clustering auxiliary losses are used. The gate is not calibrated uncertainty: it modulates regional-context injection strength rather than enforcing unconditional smoothing.

Table 1. METR-LA results over five paired mask seeds. Lower is better; the best result within each condition is in bold.

Mask	Rate	Method	MAE	RMSE
Block-T	30%	HD-TTS-AMP	$3.228 \pm .011$	$6.604 \pm .013$
		BiTGraph	$3.823 \pm .011$	$7.884 \pm .037$
		DRR-MSGF	$\mathbf{3.200 \pm .016}$	$\mathbf{6.584 \pm .049}$
	50%	HD-TTS-AMP	$3.355 \pm .013$	$6.878 \pm .022$
		BiTGraph	$3.973 \pm .018$	$8.191 \pm .046$
		DRR-MSGF	$\mathbf{3.308 \pm .020}$	$\mathbf{6.815 \pm .055}$
	70%	HD-TTS-AMP	$3.465 \pm .010$	$\mathbf{7.093 \pm .042}$
		BiTGraph	$4.136 \pm .013$	$8.481 \pm .031$
		DRR-MSGF	$\mathbf{3.446 \pm .010}$	$7.109 \pm .038$
Point	30%	HD-TTS-AMP	$3.105 \pm .007$	$6.359 \pm .015$
		BiTGraph	$3.637 \pm .012$	$7.526 \pm .038$
		DRR-MSGF	$\mathbf{3.076 \pm .012}$	$\mathbf{6.309 \pm .045}$
	50%	HD-TTS-AMP	$3.160 \pm .011$	$6.476 \pm .022$
		BiTGraph	$3.694 \pm .005$	$7.658 \pm .029$
		DRR-MSGF	$\mathbf{3.141 \pm .007}$	$\mathbf{6.456 \pm .029}$

3. EXPERIMENTS

3.1. Settings

We evaluate METR-LA traffic speeds and AQI36 air quality, forecasting 12 steps from 24 histories. METR-LA Point and Block-T masks affect only history; valid, unmasked raw targets are evaluated. Point simulates independent loss and Block-T contiguous outages. AQI36 retains its native validity mask with optional Point masks. Conditions use five paired mask seeds and shared input masks. Training uses only Eq. (18). We use $k = m = 8$, initialize $\beta = 1$ and γ to ones, and obtain $K = 31$ and $K = 5$ connected regions for METR-LA and AQI36. HD-TTS-AMP is the main reference; BiTGraph uses the common data, mask, and evaluation interface.

3.2. Main Results

Table 1 reports METR-LA results. DRR-MSGF lowers MAE over HD-TTS-AMP in all tested Block-T and Point settings. It also improves RMSE under Block-T 30%/50% and Point 30%/50%. At Block-T 70%, DRR-MSGF still improves MAE but has slightly worse RMSE, indicating that the benefit is not cross-metric consistent under extreme missingness. Across all five conditions, DRR-MSGF wins all five paired MAE runs and two-sided paired t-tests give $p \leq 0.0379$; we make no analogous across-condition significance claim for RMSE.

Table 2. AQI36 cross-domain results over five seeds.

Condition	Method	MAE	RMSE
Native	HD-TTS-AMP	$\mathbf{30.484 \pm .585}$	$\mathbf{49.333 \pm 1.407}$
	DRR-MSGF	$30.727 \pm .239$	$49.888 \pm .929$
Native+P25	HD-TTS-AMP	$31.203 \pm .689$	50.371 ± 1.035
	DRR-MSGF	$\mathbf{31.117 \pm .416}$	$\mathbf{50.333 \pm .782}$
Native+P50	HD-TTS-AMP	$31.736 \pm .496$	$\mathbf{50.403 \pm 1.204}$
	DRR-MSGF	$\mathbf{31.645 \pm .709}$	50.527 ± 1.386

Table 3. METR-LA Block-T 50% ablation over five paired seeds.

Variant	MAE	RMSE
HD-TTS-AMP	3.355 ± 0.013	6.878 ± 0.022
Physical only	3.329 ± 0.012	6.861 ± 0.049
Observation only	3.335 ± 0.016	6.887 ± 0.059
Fused graph	3.318 ± 0.011	6.829 ± 0.046
No q_{ij} penalty	3.315 ± 0.020	6.833 ± 0.055
No regional gate	3.314 ± 0.011	6.840 ± 0.042
No readout bias	3.330 ± 0.025	6.851 ± 0.038
Full **DRR-MSGF**	$\mathbf{3.308 \pm 0.020}$	$\mathbf{6.815 \pm 0.055}$

3.3. Cross-Domain Analysis

AQI36 provides a cross-domain stress test with native missingness. Table 2 shows that DRR-MSGF is worse than HD-TTS-AMP under native missingness, improves both metrics with an additional 25% Point mask, and lowers MAE but raises RMSE with an additional 50% Point mask. The heterogeneous native missingness need not make the current observation features align with transferable relations; this is a hypothesis suggested by the result, not a causal explanation. We therefore treat AQI36 as a boundary condition rather than claiming dataset-independent gains.

3.4. Ablation and Diagnostics

Table 3 reports Block-T 50% component sensitivity. The full model is best on both metrics. Removing the co-observation score penalty, the missingness-conditioned regional gate, or the readout bias increases mean error, and using only one fine relation source is weaker than the fused graph. These mean patterns are consistent with complementary roles for co-observation-guided fusion and conditional regional context, but do not isolate causal effects of individual components.

4. CONCLUSION

DRR-MSGF fuses physical and observation relations and conditions regional messages on missingness. METR-LA results are consistent with this bias; Block-T 70% and AQI36 delimit its scope.

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